

Vaibhav Talekar

AI Specialist & Machine Learning Engineer

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PROFILE

Results-driven AI Specialist and Machine Learning Engineer with 3+ years of experience delivering production AI systems and commercial full-stack applications. MSc Artificial Intelligence with Distinction from the University of Surrey. Proven track record of building scalable SaaS products, leading cross-functional teams, and delivering cutting-edge AI solutions from research to production. Strong expertise in transformers, ensemble methods, end-to-end ML pipelines, and modern web technologies (Next.js, React, TypeScript).

EMPLOYMENT HISTORY

Freelance Software Developer & AI Automation Specialist

Independent Consultant - Business Automation & AI Solutions | Guildford

Nov 2025 - Present

- Built and launched a SaaS product for Indian vendors using **AI orchestration**, automating business workflows end-to-end with intelligent task routing and multi-model AI pipelines - currently serving **100+ active customers** and growing
- Developed and sold commercial full-stack applications to multiple business clients, generating revenue through custom automation solutions and ERP systems
- Built enterprise ERP system for PMR Industries managing multi-warehouse inventory (3 warehouses, 14+ product types), production workflows, financial tracking with 5 account types, and automated Google Drive backups
- Architected Solar Sales Management System with 15-step workflow tracker, role-based authentication, RESTful APIs using Supabase (PostgreSQL), Cloudflare R2/AWS S3 integration, and automated PDF generation
- Created AI-driven business outreach platform with custom prompt engineering for automated email generation, third-party API integration (Smartlead, OpenAI), real-time streaming search, and Vercel Blob storage

Tech Stack: Next.js 15, TypeScript, React, PostgreSQL, Prisma, Supabase, Node.js, Tailwind CSS, Recharts, AWS S3, Cloudflare R2, OpenAI API

Software Developer (AI Specialist), zzish (Marvely)

EdTech Company - AI-powered language learning platform | Guildford

Feb 2025 - Nov 2025

- Led AI backend development for multi-language speaking practice platform serving GCSE, IGCSE, and IELTS preparation
- Architected speech generation, feedback systems, and automated reporting using advanced NLP models
- Built complete V2 platform using Svelte 5, delivering 40% faster performance and enhanced error handling
- Integrated payment systems and developed scalable backend infrastructure supporting thousands of users

Subject Matter Expert (Data Science), Scaler Academy

Bengaluru

Mar 2022 - Feb 2025

- Developed computer vision modules leveraging machine learning techniques for enhanced image analysis
 - Designed machine learning models using complex algorithms to solve intricate problem domains
 - Contributed to curriculum development, integrating cutting-edge technologies to enhance student learning experience
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EDUCATION

Master of Science in Artificial Intelligence, University of Surrey, Guildford

Feb 2025 - Feb 2026 | **Awarded: Distinction** | 120 Credits | Award Date: 23 February 2026

Relevant Modules: Applied Machine Learning, Computer Vision, Natural Language Processing, AI Programming, Ethics & Regulation of AI, Research Professionalism and Innovation, Thesis

Bachelor of Technology in Information Technology, Savitribai Phule Pune University, Pune

Aug 2019 - Jul 2023 | CGPA: 9.04/10 (First Class with Distinction)

KEY ACADEMIC PROJECTS

Protein Phosphorylation Site Prediction (MSc Thesis - Research Paper Under Review)

Mar 2025 - Present | Guildford

- Achieved 82.8% accuracy using novel ensemble approach combining feature engineering with transformer models
- Implemented ESM2 transformer (Facebook's protein language model) with optimized performance
- Conducted 100+ experiments across XGBoost, deep learning, and ensemble methods

Glaucoma Detection using Deep Learning (Team Leader - 31% Contribution)

Apr 2025 - Jun 2025 | Guildford

- Led team of 4 developing automated glaucoma detection achieving 94.7% Dice score and 90% IoU
- Implemented multiple architectures (UNet, FPN, DeepLabV3+, SegFormer) with comparative analysis

Network Intrusion Detection System

Apr 2025 - Jun 2025 | Guildford

- Developed RBFN model from scratch achieving 92.37% training and 88.6% testing accuracy
- Performed hyperparameter optimization using grid search, improving accuracy to 90.58%

SKILLS

AI/ML:	Deep Learning (Expert) TensorFlow (Expert) PyTorch Transformers Computer Vision (Expert) Feature Engineering (Expert) LLMs NLP
Full-Stack:	Python JavaScript/TypeScript Next.js React Svelte Node.js PostgreSQL MongoDB
Tools/Cloud:	AWS S3 Cloudflare R2 Supabase Prisma Vercel Git Docker LangChain OpenAI API

ACHIEVEMENTS & RECOGNITION

- **MSc Artificial Intelligence with Distinction** - University of Surrey (Feb 2026)
- **Research Publication (In Progress)** - Protein Phosphorylation Site Prediction (Paper under review with positive feedback)
- **SaaS Product Launch** - Built AI orchestration platform for Indian vendors with 100+ active customers
- **DataCamp Data Scientist Professional Certificate** (Jan 2024 - Jan 2026)
- **Runner-up (1st Position)** - Barclays Tech Innovation Intercollegiate Challenge (2022)

LEADERSHIP & ACTIVITIES

- **Organizer, Code Buddies Coding Event** (Nov 2022) - Orchestrated planning and execution of coding competition
- **Volunteer, Training and Placement Cell** (Aug 2021 - Mar 2023) - Managed recruitment logistics for visiting companies

LANGUAGES

English (Highly proficient) | Hindi (Native speaker) | Marathi (Native speaker)